Nouman Rasheed – AI Developer and Researcher

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AI Developer and Researcher with **4+ years** of experience in Computer Vision, **LLM fine-tuning**, and multimodal AI systems. Proven track record in academic and industrial **R&D**, including work in fast-paced startup and challenging environments. Experienced in delivering impactful solutions in **CV, LLMS, NLP, and MLOps**, with hands-on expertise in deploying production-grade AI applications.

Professional Experience

- **PrimeGate** — Riyadh, Saudi Arabia
  AI Lead Engineer — *November 2024 – Present*
  - **Access Control System**: Led development of an access control platform integrating **face recognition**, **license plate recognition**, **person tracking**, and attribute-based person search (clothing color, age, gender).
  - **Chatbot Development**: Built an **on-premises multilingual chatbot** using fine-tuned **LLaMA-12B**, supporting both **Arabic and English**. Fine-tuned embedding models for Arabic semantic search and optimized enterprise knowledge integration.
  - **Advanced Retrieval and Query Expansion**: Implemented state-of-the-art retrieval techniques and query expansion features, deploying **multi-agents** for HR, IT, Finance, Labor, and Procurement departments.
  - **AI for Safety and Compliance**: Collaborated with **GOSI and Saudi Civil Defense** to design and develop **AI-powered video analytics** solutions for workplace safety and compliance monitoring.
  - **Visitor Management System**: Developed an internal visitor management tool leveraging AI-based face recognition for secure and seamless workplace access.
- **Ink AI** — Delaware, USA
  AI Researcher / Tech Lead — *June 2024 – Present*
  - **Tech Lead: Backend, Frontend, and Research Leadership**: Worked as a tech lead overseeing **end-to-end development**, including backend services, frontend (**Kotlin**) for **Android**, and guiding research efforts. Coordinated cross-functional collaboration to deliver **production-ready AI solutions**.
  - **Digital Ink Recognition and Synthesis R&D**: Conducted research in **digital ink recognition** with focus on **handwriting generation** using reference and target text. Trained and evaluated custom models to mimic user handwriting.
  - **Model Development and Accuracy Improvement**: Designed and trained **RNN, LSTM-based, Transformer-based, and Diffusion-based models** on IAMonDB and Brush Online datasets. Achieved an accuracy improvement from **89% to 93%** in handwriting generation.
  - **Cloud Deployment and Model Serving**: Deployed trained models on **Google Cloud Platform (GCP)**, configured training pipelines, and exposed endpoints for inference. Integrated deployed services into both **web and Android applications** for seamless user interaction.
  - **Semantic Similarity Search with Vector DBs**: Built a notebook content search system using **ChromaDB** and **OpenAI LLMs** to enable context-aware retrieval through **embedding-based similarity**.
  - **System Integration and Backend Engineering**: Integrated model APIs using **FastAPI** and **RESTful interfaces**. Supported deployment and service integration with **ReactJS** frontend and **Kotlin-based** components.
  - **MLOps and Infrastructure**: Containerized services with **Docker** and utilized **PyTorch and TensorFlow** for training and serving models in **production**.
  - **Startup Contribution**: Worked closely with the startup founded by **Rich Miner (Co-founder of Android)**, contributing to R&D and operations.
- **DEER** — Dammam, Saudi Arabia
  Senior AI Engineer — *May 2022 – November 2024*
  - **Facial Recognition and Intruder Detection Pipeline**: Developed and integrated **end-to-end pipelines** for **facial recognition** and face detection systems to support intruder detection. Optimized **real-time inference** and deployed solutions within existing security infrastructure.
  - **Advanced Detection Systems**: Developed forklift and person detection systems using **MobileNet-v2** on **Google Coral TPU** (**95% accuracy**) and **YOLOv8 with DeepSort** for real-time tracking, deployed on **Nvidia Jetson Orin**.

- **High-Performance AI Solutions**: Built facial recognition solutions (**97% accuracy** with **LFFD, FaceNet, and DeepStream**) and Saudi license plate recognition (**LPRNet, 90% accuracy**).
- **LLM Model**: Created a **healthcare chatbot** (fine-tuned **LLaMA 3.1**) and implemented **GraphRAG** for document search using **Neo4j** and **LLaMA 3.2**.
- **Arabic Speech Recognition (Mulhem Project)**: Contributed to the development of **Mulhem, the first Arabic ASR system**, by improving tokenizer accuracy to **78%**.

- **Ciena (Contractor; in collaboration with Meta)** — Remote, Canada
  Contract Computer Vision Engineer — *Sep 2024 – Aug 2025*
  - **AI-Based Server Rack Scanning Platform**: Built an end-to-end CV system to analyze network chassis images and detect chassis, cards, ports, and fiber anomalies; exposed as a Flask API with Redis-backed async batching and Basic Auth.
  - **Multi-Model Inference Pipeline**: Integrated Ultralytics YOLOv8 models for **chassis segmentation**, **multicard segmentation**, **port segmentation**, and **fiber kink/knot detection**, plus a **UNet ONNX** for fiber segmentation masks.
  - **ONNX Acceleration and Quantization**: Authored tooling to export `.pt` to ONNX and perform **dynamic/static INT8 quantization** via onnxruntime, reducing latency and memory while keeping a unified inference API.
  - **Layout-Aware Port Mapping**: Designed a JSON-driven, **layout-aware interpolation** algorithm to map detections to expected card port grids and synthesize missing ports, improving robustness under occlusion/low confidence.
  - **Fiber Kink/Knot Analysis**: Implemented detection and **bend-radius** estimation from centerlines; validated against ITU **G.652/G.657** standards and produced annotated overlays and structured outputs.
  - **Batching and Best-of-3 Selection**: Buffered 3 images per view (ports/fiber) with **best-frame selection** by detection counts; added a **top/center/bottom** batch workflow that merges data and per-stage images.
  - **OCR for Labels**: Integrated **Llama 3.2 Vision** via local API for shelf/fiber label OCR with image overlays and JSON responses.
  - **APIs, Observability, and UX**: Delivered authenticated endpoints with background threads and status polling, GPU/CPU memory hygiene, structured logging, and a lightweight HTML test tool for driving/visualizing API calls.
  - **Tech Stack**: **Python**, **PyTorch**, **Ultralytics YOLOv8**, **ONNX Runtime**, **OpenCV**, **Flask**, **Redis**, NumPy, Matplotlib, SAHI (optional), CUDA.

- **Sigma Vendors** — Islamabad, Pakistan
  Computer Vision Engineer — *Sep 2021 – Apr 2022*
  - **Edge-Based Door Entry System**: Implemented a **Raspberry Pi 3** powered door entry system for healthcare security using **Tiny YOLOv5**, enabling **real-time edge inference**.
  - **Mask Detection and People Counting**: Developed a mask detection and people counting system, achieving **95% detection accuracy**.
  - **Data Visualization and Workflows**: Built interactive dashboards with **Streamlit** and designed custom workflows for streamlined data ingestion and processing.

Projects

- **Business Intelligence Dashboard**
  Python, Django, REST API, Docker, ReactJS, MySQL
  * **ETL Pipelines and Data Integration**: Built business intelligence dashboard and ETL pipelines for ingesting data from Excel, CSV, and DOC files.
  * **Forecasting and Prescriptive Analysis**: Forecasted sales KPIs using ARIMA and utilized Gemini Pro API for prescriptive analysis to provide actionable recommendations.
  * **Interactive Dashboards**: Developed dashboards with API integration and rich data visualizations to support informed business decision-making.
- **Crypto Analyzer (SkyTrading)**
  Python, Django, API, ReactJS, Docker, MySQL
  * **Trading Data Pipelines**: Implemented ETL pipelines for data ingestion and created visualizations to analyze trading history.

* **REST API and Dashboard**: Developed REST APIs in Django with a ReactJS dashboard for trading data analysis and forecasting.

- **RealEstate Property Estimator AI**
  Python, Streamlit, TensorFlow, Pandas, Docker, AWS
  * **AI-Powered Cost Prediction**: Developed a Streamlit-based AI application to predict construction costs, enabling accurate real estate investment analysis.
  * **Geospatial Analysis**: Integrated Folium maps for dynamic location selection, proximity scoring, and travel time analysis.
  * **ROI and Investment Insights**: Implemented customizable amenity preferences for ROI calculations and visualization of nearby services to assist property investors.

Research Experience

- **University of Kotli (AJK)** Kotli (AJK), Pakistan
  Undergraduate Researcher | Supervisor: Dr. Zahid Mehmood, Ph.D *Mar 2021 – Oct 2021*
  * **Intelligent Waste Management System**: Developed a custom-trained Mask R-CNN model for detecting garbage in open areas and dustbins, achieving **94.34% accuracy** for each waste class, enhancing environmental monitoring capabilities.
  * **Dataset Creation and Labeling**: Created and meticulously labeled a custom dataset of approximately **500 images** across **3 waste classes**, establishing the foundation for accurate model training and validation.
  * **Volume Measurement Techniques**: Implemented advanced computer vision algorithms for garbage volume measurement, enabling **precise segmentation** of waste areas to optimize collection schedules and resource allocation.
  * **Web Application Development**: Deployed the trained model using **Django** framework and **REST APIs**, ensuring a robust and scalable application for real-time waste monitoring and management.

Education

- **University of Kotli (AJK)** Kotli, Pakistan
  Bachelor of Science in Computer Science | CGPA: 3.64/4.00 *Oct. 2017 – Nov. 2021*

Skills and Interests

- **Languages:** Python, JavaScript, C++, Kotlin
- **Technologies and Tools:** Pandas, NumPy, Matplotlib, Seaborn, OpenCV, PySpark, Statistical Analysis, Time Series Modeling, Big Data Analytics
- **Machine Learning and AI:** GPT-based Models, LLaMA, Qwen, Model Fine-tuning and Optimization, Multimodal AI (LLaMA 3.2 Vision, Qwen), TensorFlow, PyTorch, Keras, Hugging Face, LangChain, LlamaIndex, GraphRAG
- **Backend Development:** REST APIs, FastAPI, Flask, Django, Node.js (Express.js, Mongoose)
- **DevOps and Cloud:** AWS (S3, Lambda, SageMaker, AWS Wrangler), Docker
- **Vector Databases:** Chroma, Qdrant, Pinecone, FAISS
- **Graph Databases:** Neo4j

Certifications

- **Neural Networks and Deep Learning | DeepLearning.AI** **Jul 2024**
- **Build Basic Generative Adversarial Networks (GANs) | DeepLearning.AI** **Jan 2024**
- **Introduction to TensorFlow for AI, ML, and Deep Learning | DeepLearning.AI** **Sep 2023**
- **What is Data Science? | Coursera** **Jul 2023**
- **Ethics in the Age of Generative AI | LinkedIn** **Jul 2023**
- **Ask Questions to Make Data-Driven Decisions | Google** **Jan 2023**
- **Data Manipulation with Pandas | DataCamp** **Dec 2022**
- **Foundations: Data, Data, Everywhere | Google** **Dec 2022**

- **Intermediate Python | DataCamp** **Dec 2022**
- **Introduction to Programming Using Python | DataCamp** **Dec 2022**
- **Data Visualization | Kaggle** **Apr 2022**
- **Intermediate Machine Learning | Kaggle** **Mar 2021**
- **Intro to Machine Learning | Kaggle** **Mar 2021**
- **Machine Learning | Stanford University** **Mar 2021**
- **Microsoft Technology Associate: Introduction to Programming Using Python** **Jan 2021**

Extracurricular Activities

- **CENTAIC – National Aerospace and Technology Park (NASTP)** June 2023 – Aug 2023
  AI Education Lead, Mentorship Program *Rawalpindi, Pakistan*
  - **Technical Instruction**: Developed and delivered comprehensive 3-month curriculum on Machine Learning and Deep Learning fundamentals to a cohort of interns, covering both theoretical foundations and practical implementations.
  - **Mathematical Foundations**: Guided students through essential mathematical concepts underpinning ML/DL algorithms, including linear algebra, calculus, probability theory, and statistical methods.
  - **Hands-on Programming**: Facilitated intensive PyTorch workshops, enabling participants to implement neural networks from scratch and deploy working AI solutions.
  - **Knowledge Transfer**: Bridged theoretical knowledge with practical applications, empowering the next generation of AI practitioners within Pakistan's developing aerospace technology sector.

Achievements & Awards

- **Industrial Exhibition (MUST)** 2021
  Control Automotive and Robotics Lab *Participation Shield*
  - **Recognition**: Honored as the **first** student team from University of Kotli (AJK) to present our Intelligent Waste Management System at this prestigious event, receiving a participation shield for demonstrating innovative application of computer vision techniques.